

Module 6

Computer Storage

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Module 6 — Lecture Overview

Six sections — from storage fundamentals through hardware types, cloud storage, enterprise options, memory relationships, and IoT privacy:

A

Storage and Memory — Fundamentals

Storage vs memory · Storage medium · Writing & reading · Capacity · Access time · Transfer rate

B

Storage Hardware — HDD, SSD, Flash & Optical

Hard disk internals · Platters · Tracks & sectors · Read/write heads · SSDs · Memory cards · USB flash · Optical discs

C

Cloud Storage — Concepts, Options & Security

Cloud architecture · Front/back end · Cloud services (IaaS/PaaS/SaaS/STaaS) · Selecting a provider · Securing cloud data

D

Enterprise & Other Storage Options

Data centres · RAID mirroring & striping · NAS · SAN · Tape · Magnetic stripe cards · Smart cards · RFID · NFC

E

How Memory Relates to Storage

Memory categories · Byte addressing · RAM · ROM · Flash memory · CMOS · Cache · Saving vs storing

F

Internet of Things (IoT) & Privacy

IoT devices · Smart metres · Wearables · Retail beacons · Vehicle black boxes · Privacy risks · Data consent

Storage & Memory Fundamentals

Storage vs memory · Writing & reading · Capacity · Access time · Transfer rate

Storage & Memory

Core Principle: Computers and devices use BOTH storage and memory to access and save data. Memory consists of electronic components that store instructions waiting to be executed, data needed by those instructions, and the results of processing. Storage refers to long-term, permanent access to data. (Campbell et al., 2023)

MEMORY (Volatile / Temporary)

Definition	Electronic components that store instructions, data, and results TEMPORARILY during processing.
Volatile?	YES — volatile. Contents are LOST immediately when power is removed.
Speed	Very fast — nanoseconds (ns). CPU accesses RAM in ~60 ns.
Capacity	Smaller: 4 GB to 64 GB typically. Holds only what is actively being used.
Analogy	Your DESK — holds only what you are working on RIGHT NOW. Cleared when you leave.
Examples	RAM (main memory) · CPU cache · CPU registers

STORAGE (Non-Volatile / Permanent)

Location where a computer PERMANENTLY keeps data, information, programs, and applications. Also called secondary storage.
NO — non-volatile. A storage medium RETAINS data without power.
Slower — SSD: ~100 microseconds. HDD: ~10 milliseconds. Much slower than RAM.
Larger: 256 GB to 18 TB. Holds ALL programs, files, and data permanently.
Your FILE CABINET — stores everything permanently, retrieved when needed.
(Storage hardware) HDD · SSD (internal/external) · USB flash drive · Memory card · Optical disc · Digital storage (Cloud storage)

FILE CABINET ANALOGY : Storage is similar to a file cabinet for digital content — drawers (storage) hold files permanently. You pull a folder out (load from storage → memory) to work on it, then file it back (save from memory → storage).

Storage & Memory

Core Storage Concepts

Storage Device:

The hardware that records and/or retrieves items to and from storage media. The device is the hardware (e.g. the HDD mechanism); the medium is the surface holding data (e.g. the magnetic platter).

Writing:

The process of transferring data, instructions, and information FROM MEMORY TO a storage medium. This is what happens when you press Ctrl+S (saving). Data moves from fast RAM to permanent storage.

Reading:

The process of transferring items FROM a storage medium INTO memory. Happens when you open a file or launch a program — data copies from slow storage into fast RAM for the CPU to process.



Figure 6-1 Various storage technologies

Storage vs Memory — Writing, Reading, Capacity & Access Time

Capacity is the number of bytes (characters) a storage medium can hold. A gigabyte (GB) equals approximately 1 billion bytes. A terabyte (TB) is equal to approximately 1 trillion bytes.

Table 6-1: Storage Capacity Terms (Campbell et al., 2023)

Term	Approx. Bytes	Exact Number of Bytes
Kilobyte (KB)	1 thousand	2^{10} or 1,024 e.g. Plain text
Megabyte (MB)	1 million	2^{20} or 1,048,576 e.g. MP3 song $\approx$ 3–5 MB
Gigabyte (GB)	1 billion	2^{30} or 1,073,741,824 e.g. HD movie $\approx$ 2–8 GB
Terabyte (TB)	1 trillion	2^{40} or 1,099,511,627,776 e.g. Laptop HDD: typically 1–4 TB
Petabyte (PB)	1 quadrillion	2^{50} or 1,125,899,906,842,624 e.g. Facebook stores $\sim$ 105 PB of photos
Exabyte (EB)	1 quintillion	2^{60} or 1,152,921,504,606,846,976 e.g. All internet traffic $\approx$ 4.8 EB/day
Zettabyte (ZB)	1 sextillion	2^{70} or 1,180,591,620,717,411,303,424 e.g. Global data created $\sim$ 120 ZB/year
Yottabyte (YB)	1 septillion	2^{80} or 1,208,925,819,614,629,174,706,176 e.g. Theoretical — not yet commonly used

Storage & Memory

Core Storage Concepts

Access Time:

The speed of storage devices and memory is defined by access time. Measures: (1) the time it takes a storage device to LOCATE an item on a storage medium, AND (2) the time required to DELIVER an item from memory to the processor. Lower access time = faster storage.

Transfer Rate:

The speed at which data, instructions, and information transfer to and from a device. Transfer rates for storage are stated in KBps, MBps, and GBps. Example: SATA SSD = ~550 MBps; HDD = ~120 MBps.

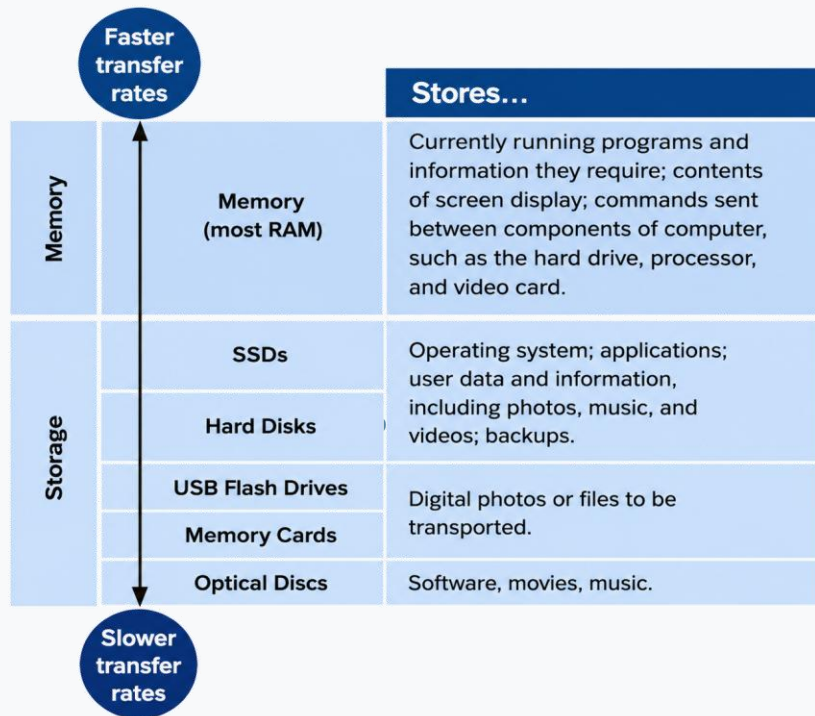


Figure 6-2 Relative speed and uses for storage media.

Storage Hardware — HDD, SSD, Flash & Optical

Hard disk internals · Platters/tracks/sectors · SSDs · USB flash · Optical discs

Storage Hardware — Hard Disk Drive (HDD)

Hard Disk (HDD): A storage device that contains one or more inflexible, circular platters that use magnetic particles to store data, instructions, and information. The MOST COMMON storage medium — desktops and laptops contain at least one hard disk. Hard drives can store data either magnetically or using solid-state storage.

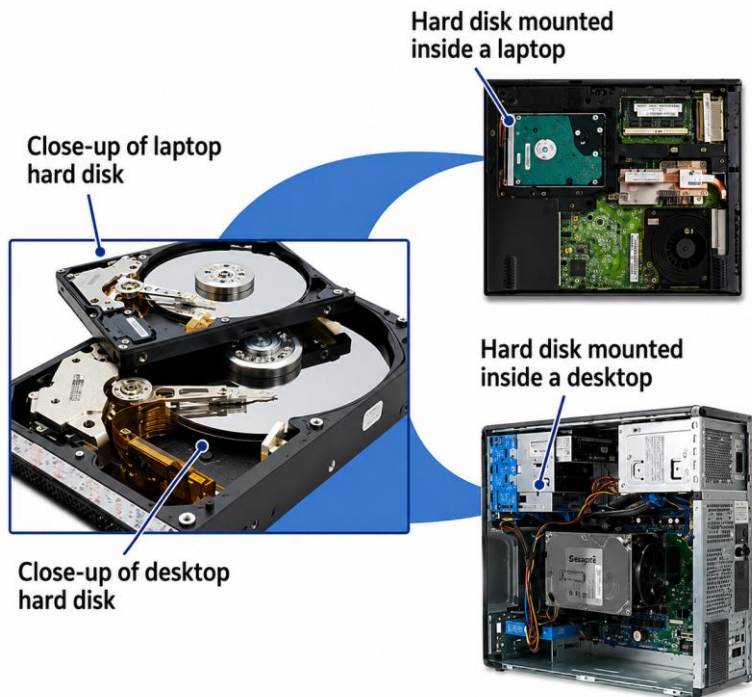


Figure 6- Typical hard disk.

Hard Drive:

The most common storage medium is the internal hard drive.

- Hard drives can store data either magnetically or using solid-state storage.
- The files stored on the internal hard drive cannot be accessed on other devices.
- Magnetic hard disk drives (HDDs) have greater storage capacity and are less expensive than their solid-state equivalents.
- The term, “hard drive”, refers collectively to hard disks and SSDs.

Platter:

The storage capacity of hard disks is determined by the number of platters that the hard disk contains, the composition of the magnetic coating on the platters, whether it uses longitudinal or perpendicular recording, and its density.

Made of aluminium, glass, or ceramic with a thin coating of alloy material allowing items to be recorded **MAGNETICALLY** on its surface.

Storage Hardware — Hard Disk Drive (HDD)

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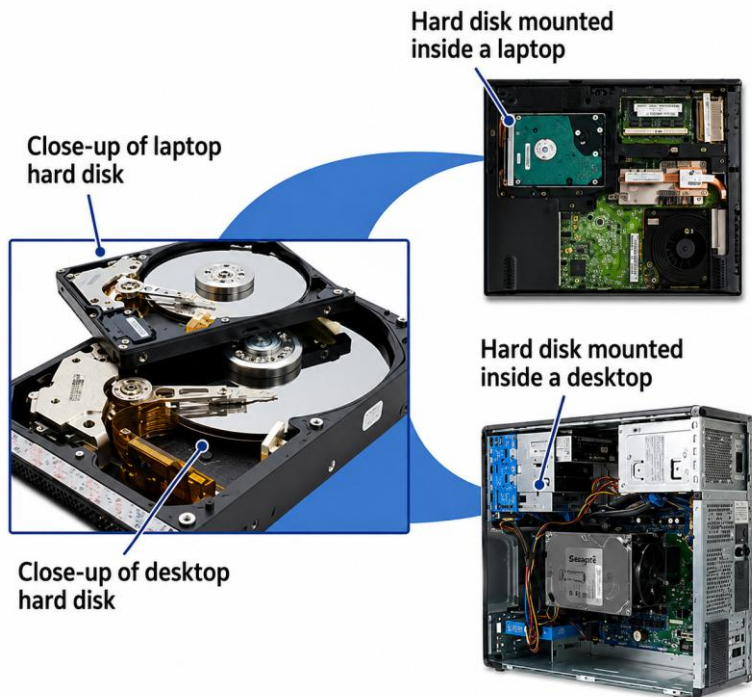


Figure 6-3 Typical hard disk.

Longitudinal Recording:

Aligns magnetic particles HORIZONTALLY around the surface. Older method with lower density.

Perpendicular Recording:

Aligns magnetic particles VERTICALLY (perpendicular to surface). Higher density = more storage capacity in same physical space. (Campbell et al., 2023)

Density:

Density is the number of bits in an area on a storage medium. A higher density means more storage capacity.

Storage Hardware — Hard Disk Drive (HDD)

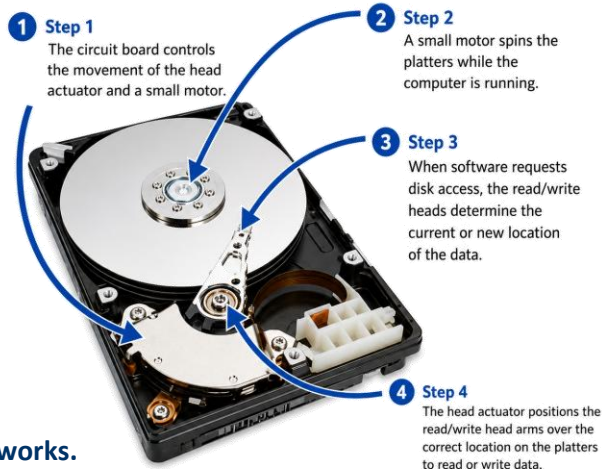
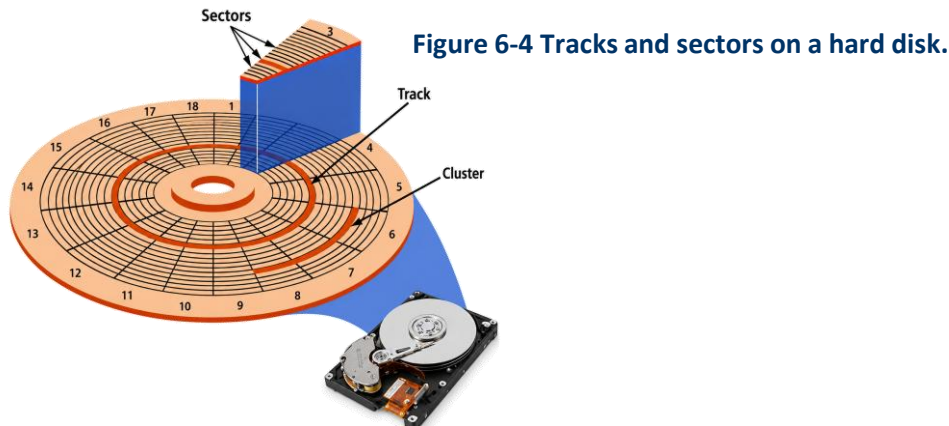


Figure 6-5 How a hard disk works.

Formatting:

The process of dividing the disk into tracks and sectors. Must be done before a disk can be used for the first time.

Track:

A track is one of the series of concentric circles on one of the surfaces of a magnetic hard disk platter. Tracks are narrow recording bands that form a full circle on the surface of the disk.

Sector:

The disk's storage locations consist of wedge-shaped sections, which break the tracks into small arcs called sectors. A sector is an individual block of data or a segment of a track.

Read/Write Head:

A typical hard disk has multiple platters. Each platter has two read/write heads, one for each side. A read/write head is the mechanism that reads items and writes items to the drive. A hard disk head actuator connects to arms that move the read/write heads to the correct location on the platter.

While the computer is running, the **platters** in the hard disk rotate at a high rate of speed. This **spinning** allows nearly instant access to all tracks and sectors on the platters. The platters continue spinning or slow down after a specified time to save power. The **spinning motion** creates a cushion of air between the platter and its read/write head. This **cushion** ensures that the read/write head floats above the platter instead of making direct contact with the platter surface.

Storage Hardware — External Drive & SSD

External Hard Drive:

- An external hard drive is a separate, free-standing storage device that connects with a cable to a USB port or other port on any device.
- Sizes and storage capacities of external hard drives vary.
- With an internal hard drive, an entire external hard drive is enclosed in an airtight, sealed case.



Figure 6-6 An external hard disk attached to a laptop.

Solid-State Drive:

- An SSD (solid-state drive) is a flash memory storage device.
- Flash memory is a type of nonvolatile memory that can be erased electronically and rewritten.
- Flash memory chips are a type of solid-state media, which means they consist entirely electronic components and contain no moving parts.

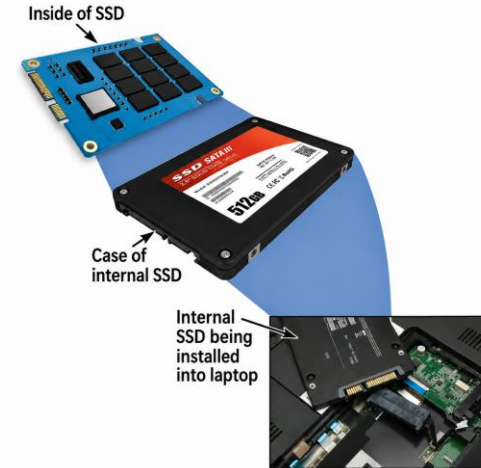


Figure 6-7 An SSD.

Storage Hardware — Memory Cards & USB Flash Drives

Flash Memory: A type of nonvolatile memory that can be erased electronically and rewritten. Flash memory chips are a type of solid-state media — they consist entirely of electronic components and contain NO MOVING PARTS.



Memory Card

Definition:

A removable flash memory storage device that enables mobile users to easily transport digital photos, music, videos, or other files to and from mobile devices and computers. A slot on a computer or device accepts multiple types of cards. (Campbell et al., 2023)

Same NAND flash technology as SSDs in a compact removable form. Types: microSD (smartphones), SD (cameras, laptops), CF (professional cameras). Speed classes: Class 10, UHS-I, UHS-II.



Figure 6-8 Memory cards often are used with cameras.



Figure 6-9 USB flash drive.



USB Flash Drive

Definition:

A removable storage device for folders and files that plugs into a USB port on a computer, making it easy to transport folders and files to other computers. Storage capacities vary. (Campbell et al., 2023)

NAND flash in a rectangular housing with a USB connector. USB 2.0: up to 60 MB/s. USB 3.0 (blue connector): up to 625 MB/s. USB-C becoming standard on newer devices. No driver installation required.

Storage Hardware — Optical Media

Optical Disc: A type of storage medium consisting of a flat, round, portable disc made of metal, plastic, and lacquer that is written and read by a LASER. Optical media include CDs, DVDs, and Blu-ray discs (BDs), but their use as storage media is DECLINING. (Campbell et al., 2023)

Table 6-2: Characteristics of Optical Disc Formats

Disc Type	Formats	Capacity	Typically Used For	How writing works
CD (Compact Disc)	CD-ROM (read-only, pressed at factory) CD-R (recordable — user writes once) CD-RW (rewritable — erases & rewrites)	700 MB	Audio CDs · Photo CDs Software distribution (legacy)	ROM: factory-pressed pits/lands. R: laser burns dye layer (permanent). RW: phase-change layer — laser reversibly alters reflectivity.
DVD (Digital Versatile Disc)	DVD-ROM (read-only) DVD-R / DVD+R (recordable) DVD-RW / DVD+RW / DVD+RAM (rewritable)	4.7 GB (single layer) 8.5 GB (dual layer)	Video · Software installation Data backup	Same principle as CD but smaller track pitch (smaller pits) due to shorter wavelength red laser (650nm vs CD's 780nm infrared). DVD+RAM allows random access like an HDD.
Blu-ray Disc (BD)	BD-ROM (read-only) BD-R (recordable) BD-RE (rewritable)	25 GB (single layer) 50 GB (dual layer) 100 GB (BDXL)	High-definition video 4K movies · Professional archiving	Blue-violet laser (405nm) — shorter wavelength = smaller beam = smaller pits = more data per disc. Highest optical capacity.

Cloud Storage — Concepts, Options & Security

Cloud architecture · Front/back end · Cloud services · Provider selection · Securing data

Cloud Storage — Architecture & Service Models

Cloud Storage: Keeps information on servers on the Internet. Home and business users choose cloud computing for ACCESSIBILITY, COST SAVING, SPACE SAVING, and SCALABILITY. Cloud computing allows companies to outsource elements of their IT infrastructure to third-party providers. In addition, cloud storage providers enable you to synchronize files, write documents, backup files on your computer or mobile device, share project work, stream music, post photos, and play games online.

Cloud Architecture — Front End & Back End:

FRONT END

Hardware and software with which the USER INTERACTS to access the cloud.

-  Laptop / Desktop PC
-  Smartphone (Android / iOS)
-  Tablet
-  Web browser (Chrome, Firefox)
-  Cloud provider's mobile app

Network
(Internet)

BACK END

Servers and storage devices that MANAGE AND STORE resources accessed by users.

-  Server racks (thousands per data centre)
-  Storage arrays (petabytes of data)
-  Security infrastructure & firewalls
-  Redundant power supplies & cooling
-  High-speed network connections

Cloud Computing Service Models

SaaS

Software as a Service
Microsoft 365, Google Docs, CBU Moodle — software delivered over internet; no local installation

PaaS

Platform as a Service
Heroku, Google App Engine — development platform in the cloud; build apps without managing servers

IaaS

Infrastructure as a Service
AWS EC2, Microsoft Azure VMs — rent virtual servers, networking, and storage

STaaS

Storage as a Service
Google Drive, OneDrive, Dropbox, AWS S3 — storage space rented over internet

DaaS

Data as a Service
Analytics platforms, open data portals — data delivered as a managed service

Desktop aaS

Desktop as a Service
Amazon WorkSpaces — full Windows/Linux desktop accessible from any device

Cloud Storage Options

- With **Google Drive** or **Microsoft OneDrive**, the files that you save are created by or can be edited or shared using the cloud-based apps that are a part of the service.
 - Other cloud storage options, such as **Box**, primarily are storage locations and do not provide corresponding apps.
 - The files are protected by security measures, including passwords and identification requirements, or encryption, and are backed up to other servers so that outages in which access is limited or denied are rare.
 - One advantage of relying on cloud-based storage is the flexibility of not having to purchase and maintain servers.
-
- Photos, songs, and videos take longer to upload than smaller text or web page files, so it is important to select a provider whose servers have sufficient bandwidth to support large file transfers.
 - Criteria to consider for **providers** include the amount of free storage offered, the cost to purchase more storage if needed, and the maximum file size that each service allows you to upload.
 - It is also important to read about the cloud storage provider's **privacy policy** and terms of agreement, to which you must consent before using the provider's services.



Figure 6-10 Cloud Storage.

Cloud Storage Options & Securing Your Data

Guidelines for Selecting a Cloud Storage Provider

- ✓ **Reputation:** Verify the company is reputable and has been in business for an extended period of time.
- ✓ **Encryption:** Choose a provider that ENCRYPTS your files — protects data in transit and at rest on servers.
- ✓ **Compatibility:** Determine whether the service is compatible with your computer(s) and mobile device(s).
- ✓ **Pricing:** Compare the price of various storage plans — choose a provider with competitive pricing.
- ✓ **File type support:** Verify the provider supports your file types. Check the MAXIMUM FILE SIZE per upload.
- ✓ **File sharing:** Choose a provider that allows you to share selected files — for collaborative work.
- ✓ **Mobile app:** Consider whether the provider offers a mobile app for access on mobile devices.
- ✓ **Bandwidth:** Photos, songs, and videos take longer to upload — choose a provider with sufficient bandwidth for large transfers.
- ✓ **Privacy policy:** Read the cloud storage provider's privacy policy and terms of agreement before using.

Securing Your Cloud Data

Choose provider carefully:

Whether using a web-based app with storage (Google Drive) or an online backup system, vet the provider's security practices and reputation first.

Read the user agreement:

Understand exactly what rights the cloud provider has over your stored data and how they may use or share it with third parties.

Advanced privacy settings:

Set the most advanced privacy settings available — restrict sharing, disable unnecessary permissions, review connected apps.

Strong passwords + 2FA:

Use strong, unique passwords AND two-factor authentication (2FA) to protect against unauthorised access from hackers.

Encrypt sensitive data:

Do NOT store sensitive data without ensuring it will be encrypted. Use end-to-end encrypted cloud services for highly sensitive files.

Enterprise & Other Storage Options

Data centres · RAID · NAS · SAN · Tape · Smart cards · RFID · NFC

Enterprise Storage — Data Centres

Enterprise Hardware: Allows large organisations to manage and store data using devices intended for HEAVY USE, MAXIMUM EFFICIENCY, and MAXIMUM AVAILABILITY. Highly available hardware is accessible 24 hours a day, 365 days a year. Includes REDUNDANCY — if one component fails, another assumes its tasks.

Enterprise Storage Centre / Data Centre

Definition: A secure location with many large computers that act as servers, making files available to users. Some organisations manage enterprise storage IN-HOUSE. Others elect to OUTSOURCE all (or at least backup) storage management to an outside organisation or a cloud storage provider.

Key features: 24x365 availability · Redundant power & cooling · Physical security (biometric entry, CCTV) · Fire suppression · Seismic protection

Zambia Context

Zambian banks (Zanaco, FNB Zambia, Standard Chartered) and telecoms (MTN, Airtel) operate their own in-house data centres for mission-critical data. Smaller organisations outsource to AWS Africa (Cape Town) or Microsoft Azure.

SMART ZAMBIA INITIATIVE: The Government of Zambia is building a National Data Centre in Lusaka to host e-government services and reduce dependency on foreign cloud providers. When complete, it will provide sovereign, highly available storage for all government ministries and national databases.



Figure 6-11 Data centre.

Enterprise Storage — RAID

RAID — Redundant Array of Independent Disks

Definition: A group of two or more integrated hard drives that may DUPLICATE DATA to improve data reliability.

RAID Level 1 — Mirroring

The SIMPLEST RAID design is Level 1, called MIRRORING, which writes data on TWO drives simultaneously to DUPLICATE the data. A Level 1 configuration enhances storage RELIABILITY.

If Disk 1 fails, Disk 2 has an exact, up-to-date duplicate — no data is lost. The system continues operating. Used by banks and hospitals for critical data.

RAID Striping

Other RAID levels use a technique called STRIPING, which SPLITS data across multiple drives. Striping IMPROVES DRIVE ACCESS TIMES — multiple drives work simultaneously.

Block A written to Disk 1; Block B to Disk 2; Block C to Disk 1; Block D to Disk 2 — reading/writing is faster because both drives work in parallel.

⚠ RAID ≠ BACKUP: RAID protects against drive FAILURE, not against accidental deletion, ransomware, or physical disaster.

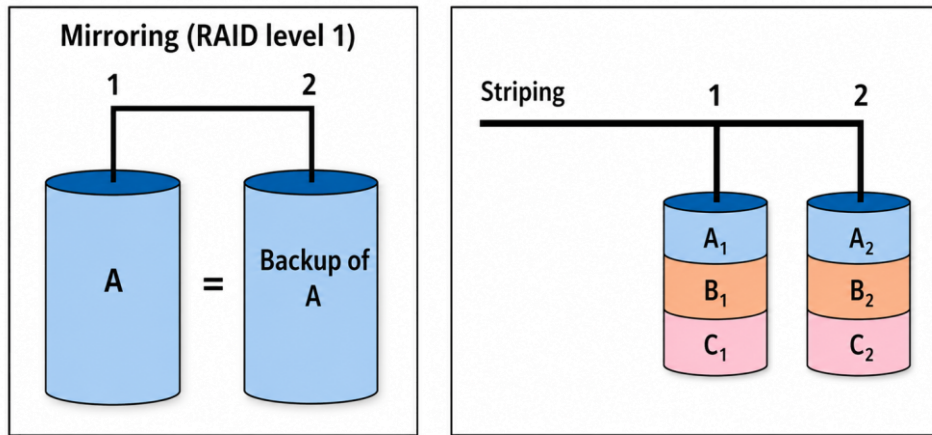


Figure 6-12 Mirroring and striping.

Other Storage Options

NAS — Network Attached Storage

A server placed on a network with the SOLE PURPOSE of providing storage to users, computers, and devices attached to the network. Has its own IP address, usually no keyboard or display, contains at least one hard drive, often in a RAID. Administrators add storage quickly by connecting a NAS to the network.

SAN — Storage Area Network

A high-speed network with the SOLE PURPOSE of providing storage to other attached servers. Contains ONLY storage devices. High-speed fibre-optic cable connects networks and servers, providing fast access to large capacities. Can connect to networks miles away.

Tape Storage

A magnetically coated ribbon of plastic capable of storing large amounts of data at LOW COST. Sequential access — must read from the beginning to reach specific data. Used primarily for long-term archival backup.

Smart Card (Chip Card)

An alternative to a magnetic stripe card that stores data on an INTEGRATED CIRCUIT EMBEDDED in the card. Two types: CONTACT (physically insert) and CONTACTLESS (tap reader). Also called chip cards. (Campbell et al., 2023)

Other Storage Options

RFID Tags

- The RFID tag consists of an antenna and a memory chip that contains the information to be transmitted via radio waves.
- An RFID reader reads the radio signals and transfers the information to a computer or computing device.
- RFID tags are either active or passive.



Figure 6-13 RFID image on a box.

NFC

- NFC is a technology (based on RFID) that uses close-range radio signals to transmit data between two NFC-enabled devices or between an NFC-enabled device and an NFC tag.
- NFC-enabled devices contain an NFC chip. NFC tags are self-adhesive so that they can be attached to any location.
- An NFC tag, similar to RFID tag, contains a chip and an antenna that contains information to be transmitted.
- Uses of NFC communications include using a mobile device to pay for goods or services, displaying a web page, making a phone call, sending a text message, or exchanging contact information.

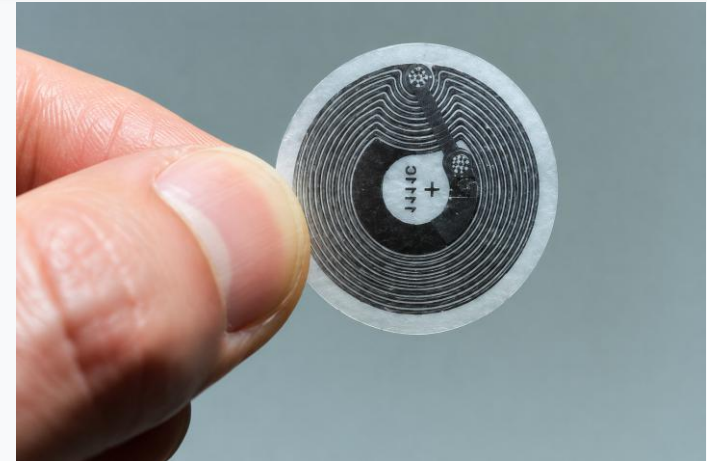


Figure 6-14 Adhesive NFC tag.

How Memory Relates to Storage

Memory categories · Byte addressing · RAM · ROM · Flash · CMOS · Cache · Saving

How Memory Relates to Storage

Memory: Consists of one or more chips on the motherboard or some other circuit board on the computer & stores three categories: (1) The operating system and programs that control/maintain the computer · (2) Applications carrying out specific tasks, such as word processing · (3) Data being processed by the applications and the resulting information.

Byte Addressing

Byte: The basic storage unit in memory. When an application's instructions and data transfer to memory from storage, they exist as bytes.

Address: A unique number identifying the location of a byte in memory. Each byte resides temporarily in a location with an address. To access data, the computer references the addresses containing bytes.

ANALOGY : Similar to seats in a stadium — one location (seat) holds a single byte (person), or can be empty.

Memory Types

RAM — Random Access Memory:

- Memory capacity affects the device's operation speed. RAM is the most common type of volatile memory.
- RAM, also called main memory, consists of memory chips that can be read from and written to by the processor and other devices. RAM can accommodate multiple programs and applications simultaneously.
- Saving is the process of copying data, instructions, and information from RAM to a storage device or to the cloud.
- Today's computers improve their processing times with cache (pronounced cash), which is a temporary storage area.
- Memory cache helps speed up the processes of the computer because it stores frequently used instructions and data.

How Memory Relates to Storage

Memory Types

RAM — Read Only Memory:

- Refers to memory chips storing permanent data and instructions. The data on most ROM chips cannot be modified so it is named as read-only.
- Manufacturers of ROM chips often record data, instructions, or information on firmware chips when the chips are manufactured.
- These chips contain permanently written data, instructions, or information, such as a computer or mobile device's start-up instructions.

CMOS — Complementary Metal-Oxide Semiconductor:

- Some RAM chips, flash memory chips, and other memory chips use **CMOS (complementary metal-oxide semiconductor)** technology as it provides high speeds and consumes little power.
- CMOS technology uses battery power to retain information even when the power to the computer is off.
- Battery-backed CMOS memory chips.
- The flash memory chips that store a computer's startup information often use CMOS technology.

Memory Types

Flash — Flash Memory:

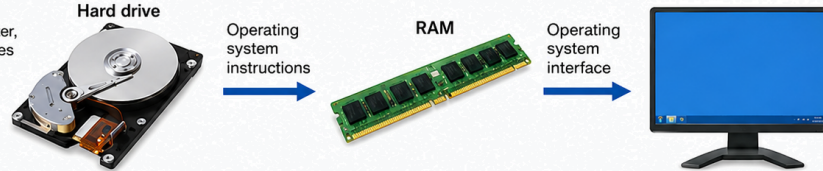
- Flash memory is a type of non-volatile memory that can be erased electronically and rewritten.
- Most computers use flash memory to hold their start-up instructions. Flash memory chips also store data and programs on many mobile devices and peripheral devices, such as smartphones, portable media players, printers, digital cameras, automotive devices, and digital voice recorders.
- Some portable media players store music on flash memory chips; others store music on tiny hard drives or **memory cards**.
- **Memory cards** contain flash memory on a removable device.

How Memory Relates to Storage

When you open an application, its instructions move from storage into RAM. The CPU fetches instructions from RAM, executes them, and writes results back — illustrating the complete Storage ↔ Memory ↔ CPU cycle.

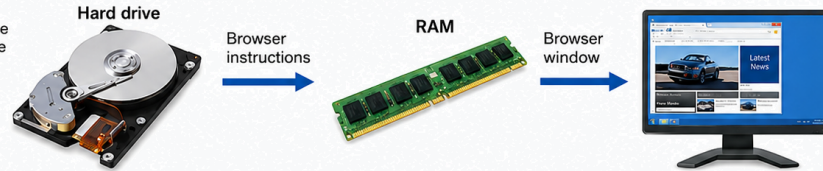
Step 1

When you start the computer, certain operating system files are loaded into RAM from the hard drive. The operating system displays the user interface on the screen.



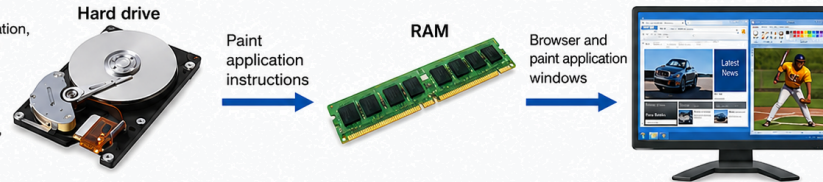
Step 2

When you run a browser, the application's instructions are loaded into RAM from the hard drive. The browser and certain operating system instructions are in RAM. The browser window appears on the screen.



Step 3

When you run a paint application, the application's instructions are loaded into RAM from the hard drive. The paint application, along with the browser and certain operating system instructions, are in RAM. The paint application window appears on the screen.



Step 4

When you exit an application, such as the browser, its instructions are removed from RAM. The browser is no longer displayed on the screen.

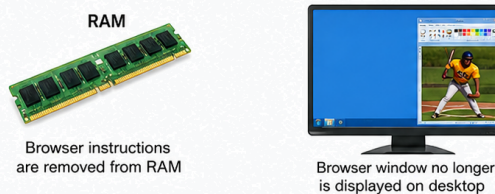


Figure 6-16 How program instructions transfer in and out of RAM.

Ethics and Issues: Internet of Things (IoT) & Privacy

IoT devices · Smart metres · Wearables · Beacons · Privacy risks · Data consent

Internet of Things — What It Is and How It Affects Us

IoT — Internet of Things: Computers and mobile devices are NOT the only items that connect to the internet. Every day, smart electric metres, wearable technology, and vehicles' black boxes submit data about us as part of the IoT. (Campbell et al., 2023)

Smart Thermostat

A temperature sensor that can send and receive data — can be set up to allow adjustment of home temperature from anywhere using a smartphone app.

Smart Medicine Bottle

A wireless chip attached to a medicine bottle can send text messages to remind you to take your medication.

Smart Trash Cans

Have sensors that monitor the amount of trash deposited and send a message notifying owners when containers need to be emptied.

Wearable Technology

Smart watches and wristbands can track pulse and heart rate, accept calls, and display notifications from a smartphone.

Retail Beacons

Devices that send LOW-ENERGY BLUETOOTH SIGNALS to nearby smartphones, alerting customers who use a payment app (such as PayPal) to personalised offers in their stores.

Vehicle Black Boxes

Vehicles submit data about us as part of the IoT. Black boxes record speed, braking, location, and driving behaviour — used in insurance telematics and accident reconstruction.

IoT Privacy Risks, Security & Data Consent

Privacy & Security Concerns: Privacy and security concerns abound with the IoT. These devices could have DEFAULT SETTINGS that PREVENT the sharing of data until obtaining the consumer's CONSENT. Companies bear the RESPONSIBILITY of ensuring sensitive data being collected is kept secure and confidential. (Campbell et al., 2023)

⚠ Continuous Data Collection

IoT devices collect data 24/7 without users being actively aware. Smart metres know when you are home. Wearables track location and health status continuously. This data profile is highly personal.

Example: A smart watch's sleep tracking data reveals your exact home schedule — valuable to burglars if data is breached.

⚠ Corporate Data Responsibility

Companies bear the RESPONSIBILITY of ensuring sensitive data being collected is kept SECURE and CONFIDENTIAL. When IoT data is breached, the consequences affect real people.

Example: A breach of wearable health data could expose HIV status, mental health conditions, or fertility information — deeply sensitive in Zambia's social context.

⚠ Default Privacy Settings

Many IoT devices ship with PERMISSIVE default settings — sharing data broadly rather than minimally. Campbell et al. (2023) note devices 'could have default settings that PREVENT sharing until consent is obtained' — this is the ideal that is often NOT implemented.

Example: A new smart thermostat may default to sharing data with third-party advertisers unless the user explicitly opts out — deep in a settings menu most users never find.

⚠ Security Vulnerabilities

Many IoT devices have poor security — default passwords, unencrypted transmission, no software updates. A hacked smart device becomes a point of entry into your home network.

Example: In 2016, the Mirai botnet infected 600,000 IoT devices (cameras, routers) with default passwords and used them to conduct the largest DDoS attack in internet history at the time.

Campbell et al. (2023): Technology experts believe that the SECURITY, HEALTH, and PRODUCTIVITY BENEFITS of IoT technology OUTWEIGH the potential risks — IF data is handled responsibly with consent and encryption.

Tutorial Discussion Activity – Module 6

Groups of 3–4 · 15 minutes group work ·

1

A Zambian software developer at a Lusaka tech startup has a computer with 8GB RAM and a 512GB SSD. (a) Explain the difference between the RAM and SSD using the volatile/non-volatile distinction. (b) They are writing a large Python data analysis script and have not saved for 1 hour when ZESCO cuts power. What data is lost? What is retained? (c) After the outage, their computer fails to boot and shows a CMOS error. What component has failed and what is the likely cause in Zambia's climate?

Storage vs Memory

2

A CBU student has an old laptop that takes 3 minutes to boot, freezes when opening 3 browser tabs, and has a 500GB HDD with only 20GB free. (a) Explain exactly WHY the HDD causes slow booting (reference access time and transfer rate). (b) Recommend a specific storage upgrade: state the type (HDD/SSD/NVMe), interface, capacity, and estimate the cost in ZMW. (c) After the upgrade, the student's friend hands them a USB flash drive with an important file. List THREE specific security risks of accepting this USB drive and THREE countermeasures.

Storage Hardware

3

A Zambian hospital in Ndola wants to store patient medical records. Requirements: 24/7 availability, protection against single drive failure, accessible from any ward, and compliant with data protection law. (a) Recommend whether they should use NAS or SAN and justify using definitions. (b) Should they implement RAID mirroring or striping? Which RAID level is most appropriate and why? (c) Should they use cloud storage? Apply at least 4 cloud provider selection criteria to evaluate suitability for sensitive patient data.

Cloud & RAID

4

A retail company is planning to deploy IoT retail beacons in all their 15 stores across Zambia to send personalised offers to shoppers. (a) Define what retail beacons are and how they work. (b) Identify THREE privacy risks this system poses for Zambian shoppers, referencing the data consent principle (c) As the ICT consultant, recommend THREE specific technical and policy safeguards the company must implement to comply with Zambia's Data Protection Act 2021.

IoT & Privacy

Practice Questions — Module 6

Prepare these for Quiz 6 and examination.

Storage Fundamentals

- (a) Define storage medium and list FIVE examples.
- (b) Distinguish 'writing' from 'reading' in the context of storage devices. Give a real-world example of each.
- (c) List ALL EIGHT storage terms from Table 6-1 in order with their approximate number of bytes.
- (d) What is the difference between access time and transfer rate? What units are used for each?

Storage Hardware

- (a) Describe the internal components of an HDD: platter, track, sector, cluster, read/write head, and air cushion.
- (b) Distinguish longitudinal recording from perpendicular recording. Which gives higher density?
- (c) Complete Table 6-2 from memory: disc types, all format variants (ROM/R/RW), capacities, and typical uses.
- (d) A student's laptop has an HDD. List THREE specific disadvantages of HDD vs SSD for a Zambian student.

Cloud & Enterprise Storage

- (a) Define cloud computing's front end and back end.
- (b) List at least FIVE of the nine guidelines for selecting a cloud storage provider.
- (c) Define RAID. Distinguish Level 1 mirroring from striping — which improves reliability? Which improves speed?
- (d) Distinguish NAS from SAN using definitions. Give a Zambian example of each.

Memory & IoT

- (a) List the THREE categories of items that memory stores.
- (b) Define CMOS technology. Why does it require a battery? What does it retain when the computer is off?
- (c) Define cache memory. Name the three cache levels (L1, L2, L3) and explain the speed difference.
- (d) Describe THREE IoT device examples. What is the company's responsibility regarding IoT data?